Hydraulic System Maintenance Report

Generated: 2026-04-15 23:24:41

Executive Summary

Total cycles analysed: 5

Cycles requiring attention: 4

Healthy cycles: 1

Overall system status: ACTION REQUIRED

System Health Overview

Pump condition:	**Severe leakage**
Accumulator condition:	**Optimal pressure**

Cycle 2 - Detailed Finding

Pump Leakage Probabilities:

Accumulator Condition Probabilities:

Top Contributing Sensors:

[+] System Efficiency (Average) - high impact (value: 57.6486)

[+] System Efficiency (Variability) - high impact (value: 22.8378)

[+] Volume Flow 1 (Average) - high impact (value: 6.3483)

AI Maintenance Finding:

FINDING:
Weak pump leakage has been confirmed, indicating a loss of fluid pressure which will reduce system efficiency.

SENSOR EVIDENCE:
The system efficiency sensor has detected high variability in pump operation, which could be an indicator of inefficient hydraulic fluid flow. Similarly, the system efficiency sensor also reports a high maximum efficiency reading, which suggests that the pump is not operating at its optimal capacity. Additionally, the volume flow 1 sensor indicates a possible restriction in fluid flow, contributing to the increased risk.

CONSEQUENCE OF DEFERRAL:
Continuing to operate the system with weak pump leakage will lead to a decline in overall system efficiency, potentially causing other equipment damage or unexpected shutdowns.

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RECOMMENDED ACTION:
Inspect and repair or replace the pump seal to address the weak pump leakage as soon as possible - within 24 hours.

Cycle 3 - Detailed Finding

Pump Leakage Probabilities:

Accumulator Condition Probabilities:

Top Contributing Sensors:

[+] System Efficiency (Average) - high impact (value: 56.6581)
[+] System Efficiency (Variability) - high impact (value: 23.3455)
[+] Volume Flow 1 (Average) - high impact (value: 6.3091)

AI Maintenance Finding:

FINDING:
Weak leakage has been confirmed in the hydraulic pump, indicating a need for immediate attention to prevent further damage or contamination of the system.

SENSOR EVIDENCE:
The system efficiency sensors flag that the pump is not operating at an average of 56.6581, which is lower than expected, and the variability is also 23.3455, higher than usual. This indicates the pump may be working harder than it needs to be and might not be providing a consistent flow. Additionally, volume flow sensor 1 indicates that the actual flow is 6.3091, which is below the expected rate.

CONSEQUENCE OF DEFERRAL:
If the weak leakage is not addressed promptly, it could lead to increased pump wear, system contamination, and potentially a shutdown.

RECOMMENDED ACTION:
Replace the hydraulic seals and inspect the pump for any damage or blockages within the next 24 hours to prevent further damage and system downtime.

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Cycle 4 - Detailed Finding

Pump Leakage Probabilities:

Accumulator Condition Probabilities:

Top Contributing Sensors:

[+] System Efficiency (Variability) - high impact (value: 22.5673)

[+] Volume Flow 1 (Average) - high impact (value: 6.3054)

[+] Volume Flow 1 (Variability) - high impact (value: 2.8525)

AI Maintenance Finding:

FINDING:
Severe leakage has been detected in the hydraulic pump, which could compromise the safe operation of the oil and gas facility.

SENSOR EVIDENCE:
This finding is primarily attributed to the system efficiency sensor, which has shown high variability in its readings (22.5673). Additionally, the volume flow 1 sensor has indicated both an average value (6.3054) and significant variability (2.8525) that increase the risk of leakage. The stable sensor has actually contributed to lowering the risk level, however its reading of 1.0 is not sufficient to counteract the other sensor indications.

CONSEQUENCE OF DEFERRAL:
If this leakage issue is not addressed promptly, it may lead to a system failure, resulting in expensive repairs and potentially affecting production.

RECOMMENDED ACTION:
The shift engineer should immediately isolate the hydraulic system to prevent further damage and notify the maintenance supervisor to schedule a thorough leak-down test and repair within the next scheduled maintenance window.

Cycle 5 - Detailed Finding

Pump Leakage Probabilities:

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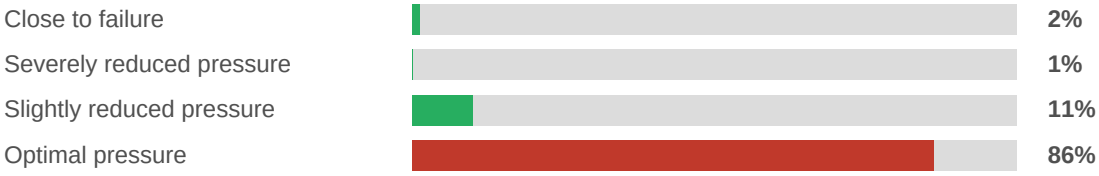
Accumulator Condition Probabilities:



Top Contributing Sensors:

[+] Volume Flow 1 (Average) - high impact (value: 3.0191)

[+] System Efficiency (Average) - high impact (value: 27.1264)

[+] Motor Power (Average) - high impact (value: 2676.6586)

AI Maintenance Finding:

FINDING:
Severe leakage from the pump was confirmed during the recent operating cycle.

SENSOR EVIDENCE:
The volume flow sensor reading of 3.0191 indicates a possible blockage in the pump or a faulty valve, causing excess fluid to bypass the system. Similarly, the system efficiency sensors (average, operating range, and maximum) flagged concerns due to high readings, suggesting inefficiencies that can further strain the system and pump. Additionally, the high motor power reading of 2676.6586 suggests an increase in resistance or an overloaded pump that's also contributing to the severe leakage.

CONSEQUENCE OF DEFERRAL:
If not addressed, the severe leakage can lead to further damage to the pump, increased energy consumption, and reduced system performance.

RECOMMENDED ACTION:
Immediate (on the next available shift): Isolate the pump and perform a thorough inspection to identify the root cause of the severe leakage, followed by a repair or replacement as necessary, to prevent further damage and ensure safe and efficient system operation.

Regulatory Reference

ISO 4413 - Hydraulic fluid power: General rules and safety requirements
API 686 - Recommended Practice for Machinery Installation and Installation Design
ISO 18435 - Diagnostics and prognostics for hydraulic systems

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All maintenance actions should be documented in the facility CMMS and approved by the maintenance superintendent before execution.